LoanLens AI

AI-Based Loan Application Processing System

LoanLens AI is an **AI-powered credit risk assessment platform** that automates loan decisioning using structured inputs **and unstructured documents (PDFs / images)**. It combines **machine learning, OCR, NLP, explainable AI (SHAP), and rule-based decisioning** to deliver fast, transparent, and auditable loan decisions.

Key Features

- Manual credit risk prediction via API
- Automatic data extraction from PDFs & images
- ML-based Probability of Default (PD) prediction
- Debt-to-Income (DTI) computation
- Explainable AI using SHAP
- Human-readable loan decision explanations
- Database logging of all predictions
- Swagger UI for easy testing

Technology Stack

Backend

- **FastAPI** – API framework
- **Uvicorn** – ASGI server

Machine Learning

- **XGBoost** – Credit risk classification
- **Scikit-learn** – Data preprocessing
- **SHAP** – Model explainability

AI & NLP

- **Tesseract OCR** – Image text extraction

- **pdfplumber / PyMuPDF** – PDF parsing
- **Transformers (DistilBERT)** – QA fallback for field extraction

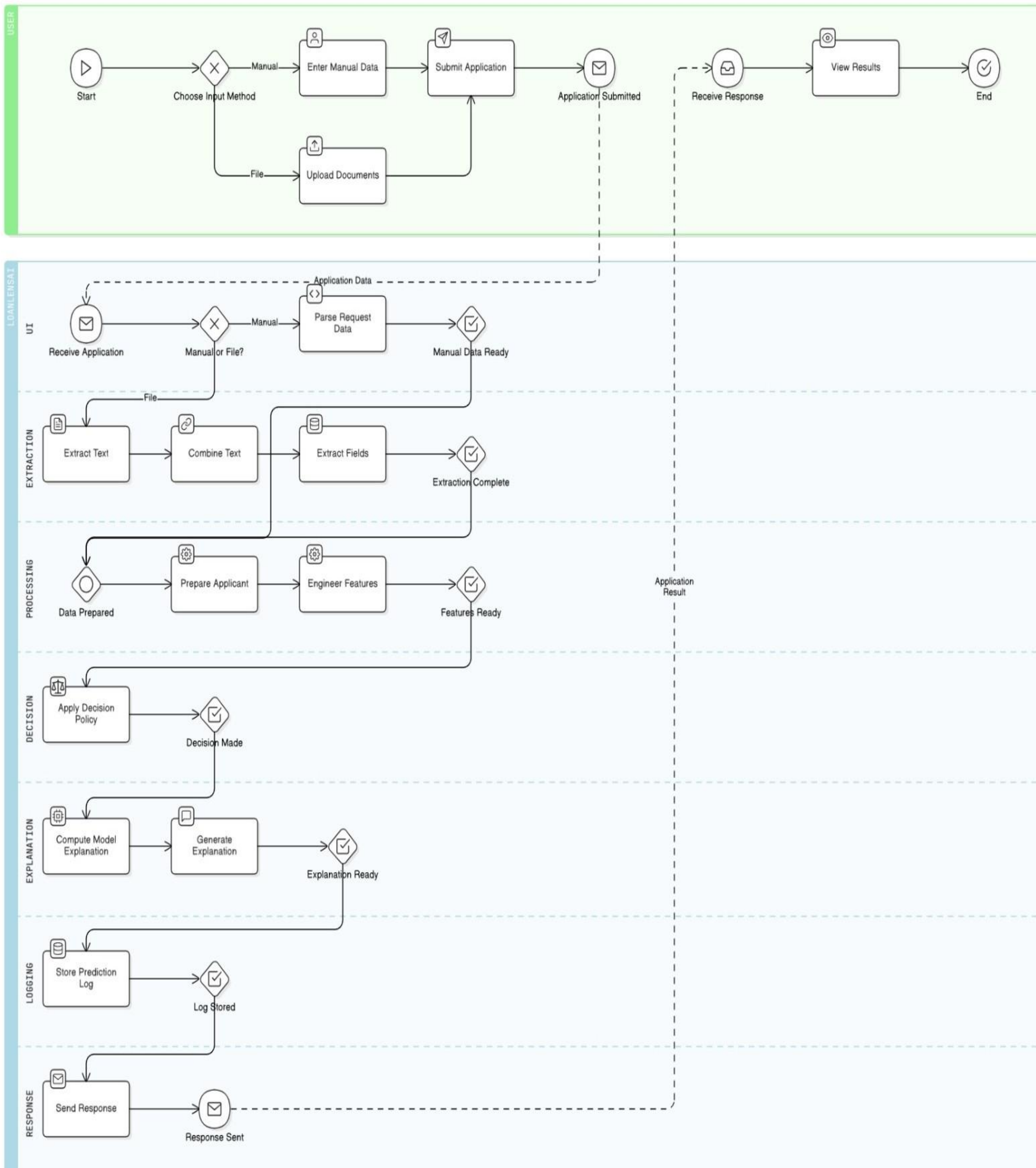
Database

- **MySQL** (via SQLAlchemy ORM)

Others

- **Pandas / NumPy**
- **LangChain (template-based explanations)**

System Architecture & Workflow



End-to-End Workflow

1️⃣ Application Startup

- Model is trained **once at startup** using the **UCI Credit Default dataset from OpenML**
- SHAP explainer is initialized

2️⃣ Input Options

Option A: Manual JSON Input

Endpoint:

POST /predict

Option B: Document-Based Input (PDF / Images)

Endpoint:

POST /predict_from_files

Process:

1. OCR extracts raw text
2. Regex + NLP extracts structured fields
3. Financial sanity checks fix OCR artifacts
4. Prediction pipeline runs automatically

3️⃣ Feature Engineering

Key engineered features:

- **Monthly Income (proxy)**
- **Debt-to-Income Ratio (DTI)**
- **Payment Delays**
- **Outstanding Credit Balances**

DTI Formula:-

DTI = (existing_monthly_debt + new_loan_emi) / monthly_income

4️⃣ Machine Learning Model

- **Model**: XGBoost Classifier
- **Output**: Probability of Default (PD %)
- **Training Data**: UCI Credit Card Default dataset (OpenML)

5️⃣ Decision Engine

Rule-based policy:

- **PD < 5%** → Approved (Low Interest)
- **5% ≤ PD ≤ 15%** → Approved (High Interest)
- **PD > 15%** → Rejected

6️⃣ Explainable AI (SHAP)

- Top risk factors extracted using SHAP
- Features ranked by absolute impact
- Improves transparency & regulatory compliance

7️⃣ Natural Language Explanation

Human-friendly explanation generated using templates:

"After reviewing factors such as payment history and outstanding balance, we are unable to approve your loan at this time."

8️⃣ Database Logging

Each prediction is stored with:

- Applicant details
- DTI & decision
- Explanation text
- Top risk factors
- Timestamp

Responsible AI & Compliance

- Transparent decision logic
- SHAP-based explainability
- No black-box approvals
- Fully auditable predictions

Future Enhancements

- Authentication & role-based access
- Model monitoring & drift detection
- LLM-based explanation refinement
- Cloud deployment (AWS / Azure)
- Integration with KYC providers

Summary

LoanLens AI demonstrates how **AI, ML, and software engineering** can be combined to build a **real-world fintech system** that is:

- Accurate
- Explainable
- Scalable
- Production-ready